AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 3)

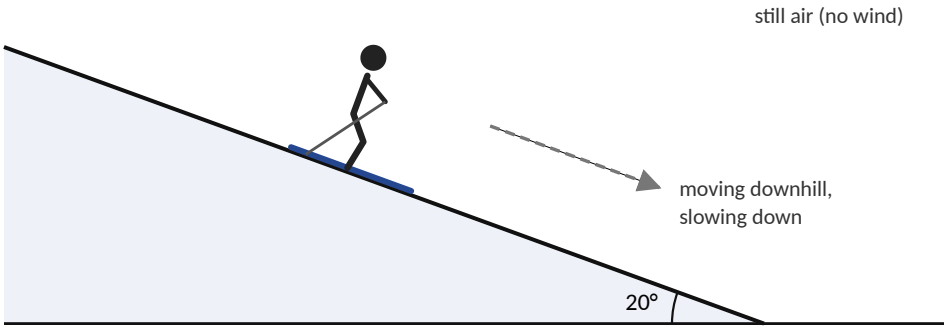
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

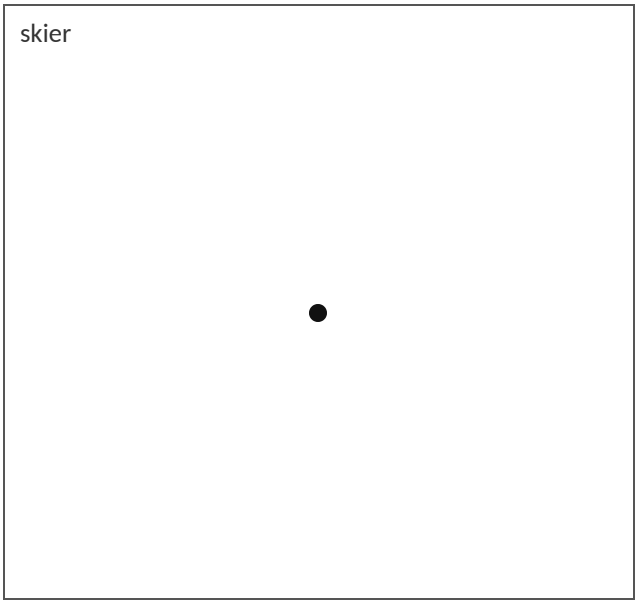
- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) A skier moves downhill on a snowy slope inclined at 20° to the horizontal. In the picture the skier moves down the slope to the right and is slowing down. Friction from the snow and drag from the air are **not** negligible.

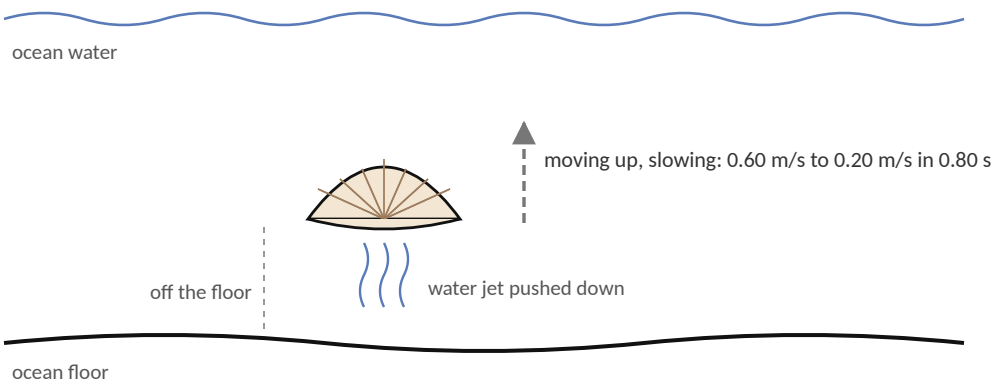


Assumptions for this practice problem: *the air is still (no wind), so the air moves past the skier opposite the skier's motion.*

Draw and label the free-body diagram of the skier. Draw the forces themselves, not components. Each force is one arrow that starts on the dot and points in the direction the force acts. No calculations.

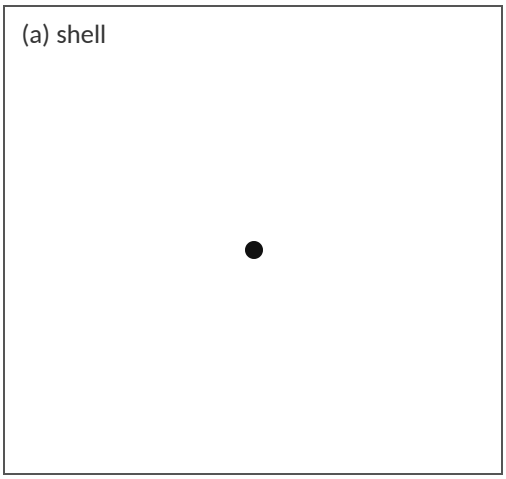


2. (25 points) A 0.320 kg shell has lifted off the ocean floor. It pushes water downward, so the water pushes it upward with a thrust of 2.00 N. After it leaves the floor, still moving upward, its speed drops from 0.60 m/s to 0.20 m/s in 0.80 s.



Assumptions for this practice problem: *water drag on the shell is negligible; its acceleration is constant; g = 9.8 m/s².*

(a) Draw the free-body diagram of the shell after it has left the floor.

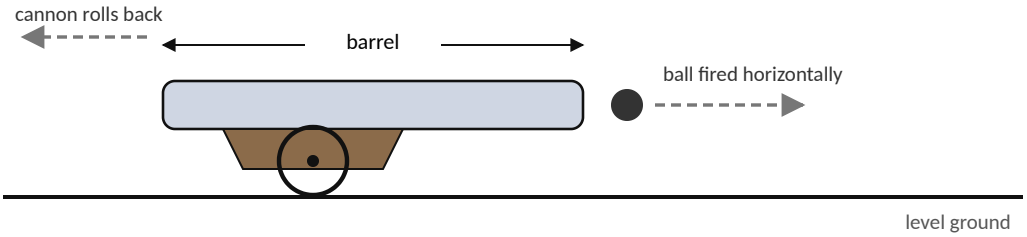


(b) Calculate the shell's acceleration, magnitude and direction.

(c) With up as positive, write Newton's second law for the shell and solve it for the buoyant force in letters.

(d) Calculate the magnitude of the buoyant force on the shell.

3. (25 points) A cannon fires a ball horizontally to the right. Find the cannon's acceleration while the ball is in the barrel.



cannonball mass	6.0 kg
cannon mass	1.5×10^3 kg
barrel length	2.46 m
distance the ball moves relative to the ground while in the barrel	2.45 m
exit speed, relative to the ground	140 m/s

The ball moves less than the barrel length relative to the ground, because the cannon slides back while the ball is in the barrel.

Assumptions for this practice problem: *the cannon starts at rest on level ground and rolls freely (ignore friction and air resistance); the ball starts from rest; the ball's acceleration in the barrel is constant; the push between the ball and the cannon is the only horizontal interaction (the mass of the gas is negligible).*

(a) Draw the free-body diagram of the cannon while the ball is in the barrel.

(a) cannon

(b) Calculate the ball's acceleration relative to the ground while it is in the barrel.

(c) Calculate the magnitude of the force the cannon exerts on the ball, and the magnitude of the force the ball exerts on the cannon.

(d) Calculate the cannon's acceleration, magnitude and direction.

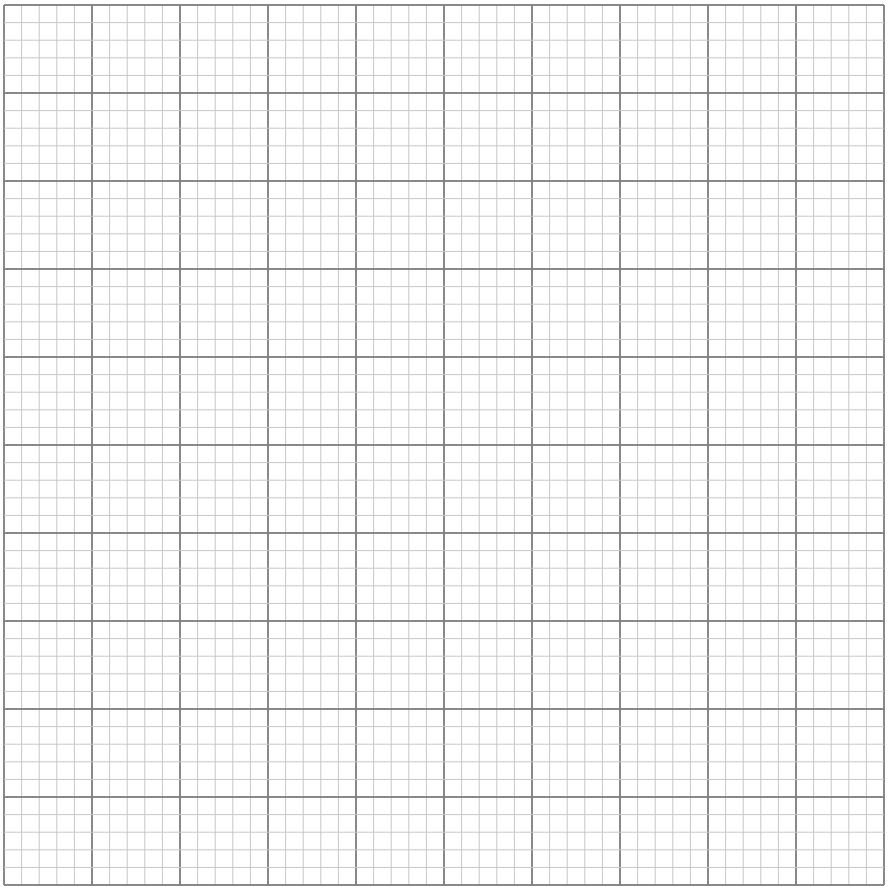
4. (25 points) Students measure the drag force F_D on a motorcycle rider in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
5.00	9.34	
9.00	30.6	
13.0	58.2	
17.0	109	
21.0	157	
25.0	220	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



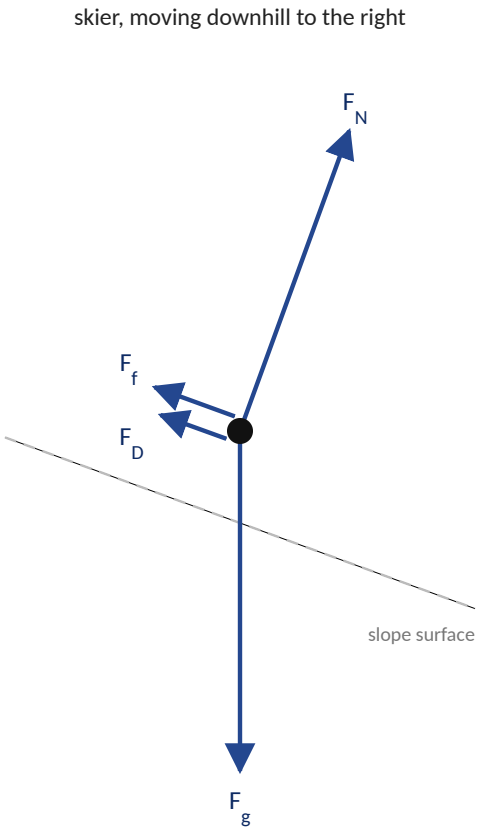
(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

Answer Key – Practice Test, Version 3 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures; answers match the significant figures of the givens.

Problems 2 and 3: diagram first, then a = ΣF/m (equation sheet), written a = F_{net}/m. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Skier free-body diagram (25 points)



Step 1 **List every force on the skier: exactly four.**

Earth pulls the skier: F_g. The snow pushes on the skier two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on the skier: drag F_D.

Step 2 **Directions.**

F_g: straight down, vertical, not perpendicular to the slope.

F_N: perpendicular to the slope, pointing out of the snow, so it tilts 20° from vertical toward the right.

F_f: along the slope, uphill (up and to the left), opposite the sliding.

F_D: opposite the skier's motion through the air. The air is still and the skier moves downhill, so drag also points uphill, parallel to F_f.

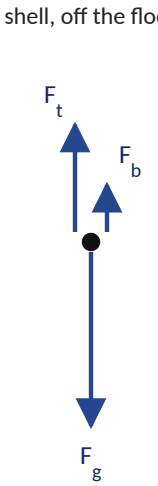
Step 3 **Check the diagram.**

Four arrows, each starting on the dot, each labeled. F_f and F_D are two separate arrows, side by side, both uphill. Speeding up, slowing down or moving at a constant speed changes the arrow lengths, not their directions.

Lengths in this key: slowing down, so the two uphill arrows together are longer than the part of gravity that pulls along the slope, and FN is a little shorter than Fg. Lengths are not graded.

Watch out: do not add the components of F_g (along and perpendicular to the slope) as extra arrows; they are not new forces. Drag is the air resistance, not a fifth force, and there is no forward "motion" force.

2. Shell: thrust and buoyancy (25 points)



Step 1 **(a) Forces on the shell after it leaves the floor.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: the shell is no longer touching the floor.

Acceleration is down (slowing while moving up), so $F_t + F_b$ is shorter than F_g.

Step 2 **(b) Acceleration from the motion data.**

Two speeds and a time: use #1. Up is +, and both velocities are upward.

$$v = v_0 + at \rightarrow a = \frac{v - v_0}{t} = \frac{0.20\ \text{m/s} - 0.60\ \text{m/s}}{0.80\ \text{s}} = -0.50\ \text{m/s}^2 \rightarrow \boxed{\textbf{0.50 m/s}^2\textbf{, downward}}$$

It is moving up but slowing down, so the acceleration points down.

Step 3 **(c) Newton's second law, up is +, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_b + F_t - F_g}{m} \rightarrow F_b + F_t - mg = ma$$

$$\boxed{\textbf{F}_b \textbf{= m(g + a) − F}_t}$$

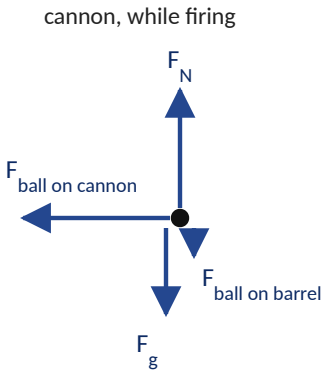
Step 4 **(d) Now the numbers, units in every parenthesis.**

$F_b = (0.320\ \text{kg})(9.8\ \text{m/s}^2 + (-0.50\ \text{m/s}^2)) - 2.00\ \text{N} = (0.320\ \text{kg})(9.3\ \text{m/s}^2) - 2.00\ \text{N} = 2.976\ \text{N} - 2.00\ \text{N} = 0.976\ \text{N}$

$\boxed{\textbf{F}_b \textbf{= 0.98 N}}$ (the subtraction leaves the hundredths place; g does not limit the significant figures).

Watch out: keep the sign of a. Using a = 0 just because the shell moves up, or dropping the thrust, gives the wrong buoyant force.

3. Cannon recoil (25 points)



Step 1 **(a) Forces on the cannon while the ball is in the barrel.**

Vertical: the ground pushes up (F_N); Earth pulls down (F_g); the ball presses down on the barrel ($F_{\text{ball on barrel}}$). Nothing accelerates vertically, so $F_N = F_g + F_{\text{ball on barrel}}$.

Horizontal: the only force is the ball pushing back on the cannon, $F_{\text{ball on cannon}}$, to the left. The vertical forces do not enter the horizontal net-force equation. Arrow lengths are not to scale; the ball's push down is tiny next to the cannon's weight.

Step 2 **Units and frames.**

All values are already in SI units.

Step 3 **(b) Kinematics for the ball.**

Use the ground-frame pair: the ball's ground displacement 2.45 m with its ground speed 140 m/s (not the 2.46 m barrel length). Time is not given: use #3.

$$v^2 = v_0^2 + 2a\Delta x \rightarrow a_{\text{ball}} = \frac{v^2}{2\Delta x} = \frac{(140\ \text{m/s})^2}{2(2.45\ \text{m})} = \frac{19\,600\ \text{m}^2/\text{s}^2}{4.90\ \text{m}} = \boxed{4.0 \times 10^3\ \text{m/s}^2}$$

Step 4 **(c) Second law on the ball, then the third law.**

$$a = \frac{F_{\text{net}}}{m} \rightarrow F_{\text{cannon on ball}} = m_{\text{ball}} a_{\text{ball}} = (6.0\ \text{kg})(4.0 \times 10^3\ \text{m/s}^2) = 2.4 \times 10^4\ \text{N} \rightarrow$$

2.4×10^4 N, forward on the ball

Third law: $F_{\text{ball on cannon}} = -F_{\text{cannon on ball}} \rightarrow$ **2.4×10^4 N, backward on the cannon**

Step 5 **(d) Second law on the cannon, horizontal only.**

$$a_{\text{cannon}} = \frac{F_{\text{ball on cannon}}}{M} = \frac{2.4 \times 10^4\ \text{N}}{1500\ \text{kg}} = \boxed{16\ \text{m/s}^2\text{, to the left (backward)}}$$

Check the frames: $t = v/a_{\text{ball}} = 0.035$ s, and the cannon moves $\frac{1}{2}(16\ \text{m/s}^2)(0.035\ \text{s})^2 = 0.0098\ \text{m} \approx 0.01$ m. Ball 2.45 m forward plus cannon 0.01 m back = the 2.46 m barrel.

Watch out: do not mix frames. The barrel length is measured on the cannon; the 140 m/s exit speed is measured from the ground, so it pairs with the 2.45 m ground displacement.

4. Drag linearization (25 points)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

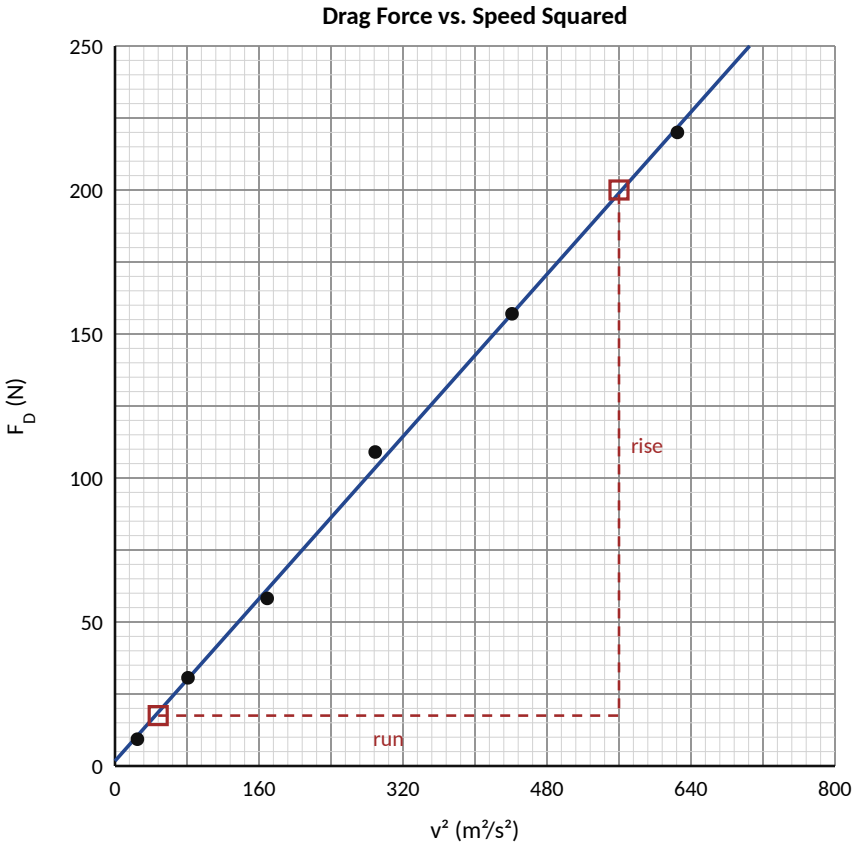
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(5.00\ \text{m/s})^2 = 25.0\ \text{m}^2/\text{s}^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
5.00	9.34	25.0
9.00	30.6	81.0
13.0	58.2	169
17.0	109	289
21.0	157	441
25.0	220	625

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

Points read off the line, away from data points: (48 m^2/s^2, 17.5 N) and (560 m^2/s^2, 200.0 N)

$$\text{slope} = \frac{200.0\ \text{N} - 17.5\ \text{N}}{560\ \text{m}^2/\text{s}^2 - 48\ \text{m}^2/\text{s}^2} = \frac{182.5\ \text{N}}{512\ \text{m}^2/\text{s}^2} = 0.3564\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.356\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.324 to 0.380 $\text{N}\cdot\text{s}^2/\text{m}^2$ from a reasonable line earns credit.

Units: $\text{N} \div (\text{m}^2/\text{s}^2) = \text{N}\cdot\text{s}^2/\text{m}^2$, and $\text{N} = \text{kg}\cdot\text{m}/\text{s}^2$, so $\text{N}\cdot\text{s}^2/\text{m}^2 = \text{kg/m}$.

1. Skier free-body diagram

Suggested partial credit	pts
F_g straight down (vertical), labeled	5
F_N perpendicular to the slope, out of the snow, labeled	5
F_f along the slope, uphill, labeled	5
F_D uphill (opposite the motion through still air), labeled, separate from F_f	5
Every arrow starts on the dot; no components and no extra forces	5
Total	**25**

2. Shell: thrust and buoyancy

Suggested partial credit	pts
(a) F_g down, thrust and buoyancy up, labeled	4
(a) No normal force	2
Acceleration: correct kinematics equation, magnitude and direction (sign)	6
$F_b + F_t - mg = ma$ written with up as +	4
Solved in letters: $F_b = m(g + a) - F_t$	3
Substitution with units in every parenthesis (and grams to kg if needed)	3
$F_b = 0.98$ N	3
Total	**25**

3. Cannon recoil

Suggested partial credit	pts
(a) F_N up, F_g down (the ball pressing down may also be shown); ball-on-cannon force backward; no forward force	5
Unit conversions (or none needed) and the matching ground-frame values	3
(b) Kinematics equation chosen and solved in letters; $a_{ball} = 4.0 \times 10^3$ m/s²	5
(c) $F = m_{ball}a_{ball} = 2.4 \times 10^4$ N, and the third law gives the same size on the cannon	5
(d) $a_{cannon} = F/M = 16$ m/s², backward	7
Total	**25**

4. Drag linearization

Suggested partial credit	pts
(a) F_D (N) on the vertical axis, v^2 (m^2/s^2) on the horizontal axis	4
(a) Slope = K	2
(b) v^2 column correct, with units	4
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line	2
(d) Two well-separated points read off the line, not data points	3
(d) K from 0.324 to 0.380, with units $N \cdot s^2/m^2$ (or kg/m)	4
Total	**25**